

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 09

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark.</i> • The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) (1)	1
01.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions (1) • Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition (1) • Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils (1) • pH / presence of other chemicals may increase or decrease the rate of degradation (1) • Adsorption onto clay / organic matter may protect from degradation and increase persistence (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
02.1	<i>1 mark for each point.</i> • Negative feedback: a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium (1) • Positive feedback: a change causes other changes that increase the original change (1)	2
02.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Increased low-level cloud: higher temperatures increase evaporation and condensation, producing more cloud; clouds have a high albedo so more sunlight is reflected and warming is reduced (2) • Increased photosynthesis: higher temperatures and CO ₂ increase photosynthesis, removing CO ₂ ; if stored in woody tissue atmospheric CO ₂ rises less (2)	4

Question	Marking guidance	Marks
02.3	1 mark for each point. - Permafrost: frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground (1) - Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming (1) - Methane hydrate: methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature (1) - If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming (1)	4
02.4	Any 3 from the points below - 1 mark each, maximum 3. - Reduced albedo as snow / ice melts (1) - Ocean acidification reducing coral survival and carbon sequestration (1) - Increased forest and peat fires releasing CO₂ (1) - Faster decomposition of soil organic matter releasing CO₂ (1) - Increased water vapour (a greenhouse gas) in warmer air (1) - Increased cirrus (high) cloud which traps infrared (1)	3
02.5	1 mark for each point. - A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable (1) - So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
03.1	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. $13 / 480 = 0.0271$ years (1) 0.0271×365 (1) = 10 days (accept 9.9-10) (1)	3
03.2	1 mark for the correct answer. $1\,340\,000 / 425 = 3153$ years (accept 3100-3200) (1)	1
03.3	1 mark for each point. - It has a very short residence time / very rapid transfer rate (480×10^{15} dm³ per year) (1) - So it moves large volumes of water each year, delivering precipitation to land; small reservoirs with fast transfers respond quickly to change (1)	2
03.4	1 mark for each point. - The ocean is so large that an added or removed volume is a tiny proportion of the total (1) - The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably (1)	2
03.5	1 mark for each point. - Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) (1) - But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.	

Question	Marking guidance	Marks
04.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Public opposition following accidents (Three Mile Island 1979, Chernobyl 1986, Fukushima 2011) (1) • Very complex, expensive technology needing an advanced industrial infrastructure (1) • Concerns about links between civil nuclear materials and weapons / terrorism (1) • Uncertainty over permanent disposal of radioactive waste (1) • Uncertain decommissioning costs (1)	3
04.2	<i>1 mark for each point.</i> • Only 0.7% of natural uranium is fissile U-235; the remaining 99.3% is non-fissile U-238 (a 'fertile' fuel) (1) • Neutron bombardment in the reactor converts U-238 to fissile plutonium-239 (via beta decays) (1) • So much more energy is harnessed from the mined uranium; fast reactors need no moderator but are complex and costly (1)	3
04.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Advantages: thorium is three times more abundant; much less radioactive waste with shorter half-lives; harder to make weapons material; no fuel enrichment needed (2) • Disadvantages: thorium-232 is not fissile so must be bred into uranium-233 (slow breeding rate); U-233 is an alpha emitter so very hazardous to handle; less developed technology with high development costs (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 192-194; AQA spec 3.3.4.1.	
05.1	<i>Correct answer only.</i> • B (1)	1
05.2	<i>1 mark for each point.</i> • Number all possible sampling sites (eg on a grid map) (1) • Select the sites using a random number generator / table; (regular shape: grid coordinates from random numbers) (1)	2
05.3	<i>1 mark for each point.</i> • Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis (1) • So the sample does not represent the whole area and conclusions are invalid (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
06.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Temperature: most chemical / biological reactions are faster at higher temperature so degradable pollutants (eg sewage) break down faster; but faster decomposition may increase deoxygenation (2) • Light: provides the activation energy for chemical reactions; photodegradation of pesticides (eg neonicotinoids break down fastest in sunlight); light drives photochemical smog reactions producing secondary pollutants (2) • Oxygen: needed for aerobic bacterial decomposition of organic wastes / sewage; oxidation of sulfide ores produces sulfur dioxide / acid mine drainage; oxidation state affects the solubility of metals (2)	6

Question	Marking guidance	Marks
06.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Pollutant ions adsorb onto (the surface of) clay particles / organic matter in soil or sediment (1) • So they are immobilised and cannot be absorbed by organisms / leached into water; radiation is absorbed in the ground (1) • But they may be released later if conditions change (eg storms disturbing sediment, low pH) (1)	2
06.3	<i>1 mark for each point.</i> • Many heavy metals (eg lead, aluminium) are more soluble at low pH (1) • So they are mobilised / leached from soil or sediments and can be absorbed by organisms / enter water bodies (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 240-242, 255; AQA spec 3.4.2.2.	
07.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Topography: a narrow exit from a large deep basin so a small dam holds a large volume (1) • Geology: impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity (1) • Catchment area: large enough with regular, reliable rainfall / river inflow; climate not too hot (evaporation losses) (1) • Catchment land use: low pollution risk (industry, pesticides); low soil erosion (sedimentation reduces capacity) (1) • Infrastructure / proximity to demand: access for construction, cost of transporting water (1) • Existing land use: value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis (1)	4
07.2	<i>Up to 3 marks in total; award marks in proportion to the completeness of each point.</i> • Advantage: fewer land-use conflicts as no land is flooded; large volume near coastal cities (up to 2) • Disadvantages: destroys / changes important intertidal habitats (mudflats for wading birds); obstacle to shipping; collects pollution from the whole river catchment; sedimentation (up to 2)	3
07.3	<i>1 mark for each point.</i> • Very energy-intensive and expensive (reverse osmosis / distillation) (1) • Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East) (1)	2
07.4	<i>1 mark.</i> • Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.	
08.1	<i>1 mark for each point.</i> • The human ear is sensitive to a very wide range of sound pressures / powers (about 10^{12} range) (1) • A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity (1)	2

Question	Marking guidance	Marks
08.2	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Difference = 30 dB = 3 x 10 dB (1) • = 10³ = 1000 times (1)	2
08.3	<i>1 mark for each point.</i> • 88 dB (1) • Doubling the sound intensity increases the level by 3 dB (logarithmic scale) (1)	2
08.4	<i>1 mark.</i> • 0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing; quieter sounds exist but cannot be heard (1)	1
08.5	<i>1 mark for each point.</i> • 20 to 20 000 Hz (1) • dB(A) weights sounds between about 1000 and 4000 Hz more heavily (1) • Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance; range narrows with age or noise damage (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.	
09.1	<i>1 mark for each point.</i> • The lowest ore purity that can be mined economically with existing technology (1) • Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves) (1)	2
09.2	<i>1 mark for each point.</i> • X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper (1) • But X is deep: costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) (1) • X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs (1) • X is far from transport infrastructure so transport costs are high (1) • Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall (1) • Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching); both sulfides risk acid mine drainage (1)	6
09.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Higher market value / price of the metal (1) • Improved extraction technology (1) • Lower energy costs (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Smoke: solid particulates (PM₁₀, PM₅, PM₁) from incomplete combustion; small particles stay suspended, penetrate lungs; toxic chemicals (acids, phenols, metals) carried on particles; high albedo cools climate; blocks light reducing photosynthesis; damages buildings • Smoke smogs form under temperature inversions in valleys (London 1952, 12 000 deaths); synergism with SO₂ • Controls: Clean Air Act 1956 (smokeless zones), smokeless fuels / gas / electricity, coal treatment, electrostatic precipitators, cyclones, bag filters, scrubbers, DPFs, better combustion; smoke smogs have largely been eliminated in UK cities • Acid rain gases: SO₂ (coal combustion, smelting sulfide ores) oxidised by ozone to SO₃ - sulfuric acid; NO_x (high temperature combustion) - nitric acid; HCl from PVC incineration; reactive gases travel long distances (transboundary pollution) • Effects: pH denatures proteins / enzymes in exposed cells (gills, eggs, root hairs, stomata); dissolves exoskeletons; leaches Ca and Mg then mobilises toxic Al and Pb; corrodes metals, dissolves limestone; severity depends on bedrock (limestone neutralises) • Controls: fuel desulfurisation (coal washing, hydrosulfurisation, low-sulfur ship fuels), dry and wet FGD, fluidised bed combustion, catalytic converters, urea sprays; switching from coal to gas / renewables • Evaluation: SO₂ emissions in Europe have fallen dramatically and lake pH is recovering; NO_x harder to control because of growing vehicle numbers; catalytic converters need lead-free fuel and warm engines; FGD is costly and produces wastes / gypsum; controlling primary pollutants also prevents secondary pollutants (ozone, PANs) • Judgement on relative success of controls for particulates vs SO₂ vs NO_x	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	Smoke is particulate matter from incomplete combustion of coal, diesel, wood and wastes. Its effects follow from particle size: PM10 and smaller particles stay suspended, travel far and penetrate the lungs, carrying acids, phenols and metals; smoke also reflects sunlight (cooling), blocks light for photosynthesis and blackens buildings. Under a temperature inversion in a valley it accumulates as smog, and it acts synergistically with sulfur dioxide - the London smog of 1952 killed about 12 000 people. Acid rain gases are different: sulfur dioxide from coal combustion and sulfide smelting, and nitrogen oxides from high-temperature combustion, are reactive and soluble, forming sulfuric and nitric acids that travel long distances across borders. Low pH denatures proteins in exposed cells such as gills, eggs and root hairs, dissolves exoskeletons and limestone, corrodes metals and leaches calcium and magnesium from soils while mobilising toxic aluminium, with severity depending on bedrock.	
	Smoke control has been the clearest success: the Clean Air Act's smokeless zones, the switch to gas and electricity, electrostatic precipitators, cyclones, bag filters and diesel particulate filters have virtually eliminated smoke smogs in developed countries. Sulfur dioxide has also fallen dramatically through fuel desulfurisation, flue-gas desulfurisation and the move from coal to gas and renewables, and lake pH is recovering, although FGD is costly and produces waste. Nitrogen oxides remain the hardest: catalytic converters and fluidised-bed combustion work, but converters need unleaded fuel and warm engines and vehicle numbers keep rising. Controls have therefore succeeded for particulates and sulfur, only partly for nitrogen.	
	Weaker answer - Level 2 (5 marks)	
	Smoke is made of small particles from burning coal and diesel. It causes breathing problems and smogs like in London in 1952. Acid rain is caused by sulfur dioxide and nitrogen oxides which dissolve in rain and make it acidic. Acid rain kills fish and trees and damages buildings. Smoke is controlled by smokeless zones and filters. Acid rain is controlled by taking the sulfur out of fuel and catalytic converters in cars. These methods have reduced pollution a lot, especially smoke.	
	What makes the difference	
	• Level 3 links each effect to a property (particle size, solubility, reactivity, pH denaturing proteins); Level 2 describes effects without the mechanism.	
	• Level 3 evaluates each control separately with evidence (recovering lake pH, cost of FGD, why NO _x is harder); Level 2 gives one general sentence.	
	• Both answers use the 1952 smog, but Level 3 explains the conditions (inversion, synergism with SO ₂) that caused it.	
10.2	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> • Reduces growth rates of trees / crops (1) • Damages leaf cuticles (allowing SO ₂ to cause more damage) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 245-254; AQA spec 3.4.3.2.1-3.4.3.2.5.	

Question	Marking guidance	Marks
11.1	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25

Question	Marking guidance	Marks
11.2	Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud • Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring • Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	25